Data Analysis Project

Project Guidelines and Grading Rubrics

Instructor: Eric Li | Semester: Fall 2026

DDL: Dec 31, 11:59 PM

Late submissions will be allowed only for excused reasons!

1. Project Overview

Students will work **individually** to:

- **Question**: Identify a real-world problem, or a series of related problems.
- **Collection**: Acquire at least one appropriate open-source dataset.
- **Cleaning**: Perform comprehensive data cleaning.
- **Analysis**: Perform EDA to map trends, distributions, and correlations.
- **Communication**: Present your findings in a formal technical report.

Remember this is an **exploratory, iterative** process:

- It's OK not to know what questions to ask at first. Keep gauging and playing around with data!
- When new patterns and relationships are discovered, be bold to add to/change your original questions.
- Be cautious about making strong conclusions and causalities!

2. Final Deliverables

Students should submit any related material no later than **Dec 31 11:59 PM** to zhengyi.li@ssbs.sh.cn.

- **Python Source Code (.ipynb / .py)**: Documented notebook/python file showing data loading, cleaning, EDA, visualizations, etc.
- **Technical Data Analysis Report (notebook / PDF)**: You can either incorporate your report within the notebook file (and potentially export as HTML or others), or write a separate PDF/word document and include plots and screenshots of your original codes.
- **Raw Datasets**: If you have .csv/Excel files (any local file) loaded within the notebook, be sure to include the datasets in the same submission package.
- **Interactive Visual Dashboard (Optional/Bonus)**: Integration with tools like Power BI, Tableau, or Streamlit to validate or expand upon the Python analysis.

3. Report Structure

Below is a recommended structure for your data analysis report:

- **Title**: Project title, name and class.
- **Table of Contents**: Include a TOC page if it's a PDF. TOC can be naturally created within notebook.
- **Introduction**: Define the scope of the problem, global/local impacts, and clear project aims.
- **Data Collection**: Data source documentation, feature identification, and target selection.
- **Data Cleaning** : Methods used for handling null values, data preprocessing, restructuring, etc.
- **Data Analysis**: Methods used for modeling and visualizing trends, distributions, and correlations, application of ML models for prediction, feature weight analysis, etc.(optional).
- **Conclusion**: Evaluation of key takeaways, challenges faced (e.g., coding limitations, data complexity), and a retrospective analysis.
- **References** : Properly cited data repositories and tool documentation.

4. Grading Rubrics

Criteria	Points	Target Expectations (A-Grade Benchmark)
Problem Definition & Scope	20	Clearly identify a question or a series of questions of concern. Your questions should be practical, potentially revealing, somewhat high-impact, global or regional issue, and at least partially answerable by analytical methods.
Data Collection	15	Source valid data from authoritative repositories. Explain how the data is related to the problem scope or subject of concern. Properly load datasets and libraries to the workspace.
Data Cleaning	15	Identify features of interest. Perform proper data cleaning, selection, restructuring, and aggregation ready for further analysis.
Data Analysis & Visualization	15	Perform statistical analysis on trends, distributions, and correlation (with potential modeling tools). Communicate through clear, good-looking, and possibly interactive visualizations.
Interpretation & Conclusion	20	Interpret the results of your analysis, and relate back to the questions posed at the beginning. Make reasonable inferences but not arbitrary conclusions. Suggest implications, limitations, and spaces for further exploration.
Formatting & References	5	Report follows a clear structure and accurate Table of Contents. Formally cite all data and external literature.
Total	90	

Let’s look at a student example.

5. Discussion

Think about:

- In what aspects is this example doing well? Doing not so well?
- Any main takeaway for your own project?
- Any room for improvement?

Project Showcase (?)

TBD

Any question?